

Muqeeth Mohammad

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SUMMARY

Software Development Engineer with 6+ years of experience building scalable full-stack applications, distributed systems, and AI-powered backends. Specialized in LLM integration, RAG pipelines, and agentic workflows using LangChain and LangGraph. Proven track record delivering high-availability systems at scale, with strong command of AWS cloud infrastructure (ECS, EC2, CloudWatch, VPC, CodePipeline), Node.js, React, Python, and microservices architecture.

EXPERIENCE

Accenture

Senior Software Development Engineer

Feb 2023 - Dec 2024

- Engineered RAG pipelines using LangChain/LangGraph with step-back prompting and multi-query generation, reducing data retrieval time by 40% for 50-100 production users.
- Designed and deployed distributed microservices architecture across 8+ services on AWS ECS with ALB and VPC configuration, improving system scalability by 60% and reducing inter-service coupling for 3 mission-critical applications.
- Constructed LLM-based evaluation frameworks integrated into CI/CD pipelines on AWS CodePipeline, cutting AI model validation time by 50% and accelerating production deployment cycles from bi-weekly to continuous delivery.
- Designed fault-tolerant backend systems leveraging AWS CloudWatch for observability and structured incident response protocols, reducing MTTR by 45% and achieving 99.95% uptime across production environments.
- Partnered directly with clients and stakeholders to gather and clarify technical requirements, manage delivery timelines, and align engineering priorities across multiple concurrent project tracks.
- Led technical design reviews and mentored a team of 4 junior and mid-level engineers, establishing coding standards, PR review processes, and architectural guidelines adopted across the engineering org.
- Architected end-to-end security for AI-integrated APIs including OAuth 2.0, JWT-based authentication, IAM role scoping on AWS, and secrets management via AWS Secrets Manager, ensuring compliance across all production services.

Software Development Engineer

Sep 2018 - Feb 2023

- Constructed and deployed scalable full-stack applications using Next.js (SSR/SSG), Node.js microservices, and Docker containerization on AWS EC2 with S3 for asset storage, sustaining 2,000+ concurrent users at 99.9% uptime.
- Designed event-driven architecture integrating Kafka for real-time data streaming and BullMQ for asynchronous job processing, handling 5,000+ messages/hour while reducing system latency by 30% across data-intensive modules.
- Implemented API contracts, caching solutions via AWS ElastiCache, and MongoDB sharding and indexing strategies, achieving a 50% reduction in API response time and significantly strengthening overall application performance.
- Developed and maintained full-stack features using React, Next.js, and Node.js/Express.js RESTful APIs, contributing to system stability through structured troubleshooting and full-stack development best practices.
- Established unit and integration testing practices using Jest and Supertest across backend services, achieving 80%+ code coverage and reducing production bug rate by 35% over two release cycles.
- Optimized frontend rendering performance through code splitting, lazy loading, and SSG strategies in Next.js, improving Lighthouse performance scores by 40% and reducing time-to-interactive by 2.5 seconds on key user flows.

PROJECTS

TheWhiskCorner - Full-Stack Web Application thewhiskcorner.com

Oct 2025 - Present

- Engineered a responsive, mobile-first e-commerce platform using Next.js and Tailwind CSS, integrating custom RESTful API route handlers to manage dynamic product catalogs and streamline client-server data delivery.
- Designed an infinite-scrolling gallery by coupling React Query and Intersection Observer API with backend pagination and server-side filtering, reducing initial database query loads by 80% and ensuring a seamless browsing experience.
- Constructed an end-to-end custom quote generation system combining multi-step React forms with secure backend POST endpoints for data validation, reducing form friction and increasing estimated conversion rates by 30%.
- Implemented full-stack authentication and role-based access control (RBAC) using encrypted sessions and tokens, securing backend API routes and dynamically rendering administrative dashboards based on real-time user privileges.

StudyMate AI - Autonomous Learning Platform studymate-sigma-six.vercel.app

Jan 2026 - Feb 2026

- Architected an agentic AI backend using LangGraph and Next.js 14, integrating Search and News APIs to eliminate LLM hallucinations and generate real-time, week-by-week learning curriculums grounded in live web data.
- Engineered a stateful, context-aware conversational agent leveraging LangGraph's Checkpointer and MongoDB, implementing asynchronous background nodes to analyze chat history and extract user learning gaps for personalized quiz generation without impacting frontend latency.

- Constructed a full-stack progression system combining GPT-4o and Zod for structured JSON output validation, integrating adaptive quiz logic into a responsive React/Tailwind CSS dashboard with custom SVG rendering and Recharts for real-time progress visualization.

SKILLS

Languages: JavaScript, TypeScript, Python
AI Technologies: Agentic Workflows, GenAI, LangChain, LangGraph, NLP, Retrieval Augmented Generation (RAG)
Frontend: React, Next.js, Redux, Tailwind CSS, UI/UX
Backend: BullMQ, Docker, Express, Kafka, LangChain, LangGraph, Node.js, REST APIs
Data & Databases: MongoDB, MySQL, Neo4j (Graph Database), Qdrant (Vector Database), Redis, NoSQL
Cloud & Infrastructure: AWS (EC2, S3, ECS, ELB, VPC, CloudWatch, CodePipeline, ElastiCache, IAM), CI/CD, Database Indexing/Sharding, Docker, Git, Microservices Architecture.

EDUCATION

Master of Science in Computer Information Systems - AI Specialization Indiana Wesleyan University, Indiana, United States	Jan 2025 - Aug 2026
Bachelor of Science in Computer Science Osmania University, India	Mar 2014 - Aug 2018